AP Calculus BC Concept Notes

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Contents

1 The Squeeze Theorem

The Squeeze Theorem

If $f(x) \leq g(x) \leq h(x)$ when x is near a (except possibly at a) and

$$\lim_{x \to a} f(x) = \lim_{x \to a} h(x) = L$$

then

$$\lim_{x \to a} g(x) = L.$$

2 Epsilon Delta Definition of a Limit

Epsilon-Delta Definition of a Limit

Let $f(x)$ be defined on some open interval (I) that contains the number a, except possibly at a itself, then we say:

$$\lim_{x \to a} f(x) = L \iff$$
$$\forall \epsilon > 0, \quad \exists \delta > 0 \text{ s.t. } 0 < |x-a| < \delta \implies |f(x) - L| < \epsilon$$

- $|f(x) - L| < \epsilon \iff L - \epsilon < f(x) < L + \epsilon$
- $|x - a| < \delta \iff a - \delta < x < a + \delta$

Epsilon-Delta Proof Example 1

Prove that $\lim_{x \to 3} 4x - 5 = 7$.
Given $\epsilon > 0$, find δ s.t. if $|x-3| < \delta$, then $|4x - 5 - 7| < \epsilon$.

$$|4x - 12| < \epsilon$$
$$\iff 4|x-3| < \epsilon$$
$$\iff |x-3| < \frac{\epsilon}{4}$$
$$\boxed{\text{Choose } \delta = \frac{\epsilon}{4}}$$

Actual Proof:
Given $\epsilon > 0$, let $\delta = \frac{\epsilon}{4}$, we want to show if $|x-3| < \delta$, then $|4x - 5 - 7| < \epsilon$.

$$|x-3| < \frac{\epsilon}{4}$$
$$\iff 4|x-3| < \epsilon$$
$$\iff |4x - 12| < \epsilon$$
$$\iff |4x - 5 - 7| < \epsilon$$

By the $\epsilon - \delta$ definition:

$$\boxed{\lim_{x \to 3} (4x - 5) = 7}$$

Epsilon-Delta Proof Example 2

Use the $\epsilon - \delta$ definition to prove that $\lim_{x \to 9} (\sqrt{x} + 2) = 5$.
Given $\epsilon > 0$, find δ s.t. if $|x - 9| < \delta$, then $|\sqrt{x} + 2 - 5| < \epsilon$.

$$|\sqrt{x} - 3| < \epsilon$$
$$\iff |\sqrt{x} - 3| \cdot |\sqrt{x} + 3| < \epsilon|\sqrt{x} + 3|$$
$$\iff |x - 9| < \epsilon|\sqrt{x} + 3|$$

Let $\delta < 1$.

$$8 < x < 10$$
$$2\sqrt{2} < \sqrt{x} < \sqrt{10}$$

$$2\sqrt{2}+3 < \sqrt{x}+3 < \sqrt{10}+3$$
$$2\sqrt{2}+3 < |\sqrt{x}+3| < \sqrt{10}+3$$
$$|x-9| < \epsilon|\sqrt{x}+3| < (2\sqrt{2}+3)\epsilon$$
$$\boxed{\text{Choose } \delta = \min\{1, (2\sqrt{2}+3)\epsilon\}}$$

Given $\epsilon > 0$, let $\delta = (2\sqrt{2}+3)\epsilon$.

$$|x-9| < (2\sqrt{2}+3)\epsilon$$
$$\iff \frac{|x-9|}{|\sqrt{x}+3|} < \frac{(2\sqrt{2}+3)\epsilon}{|\sqrt{x}+3|} < \frac{(2\sqrt{2}+3)\epsilon}{2\sqrt{2}+3}$$
$$\iff |\sqrt{x}-3| < \epsilon$$
$$\iff |\sqrt{x}+2-5| < \epsilon$$

By the $\epsilon - \delta$ definition:

$$\boxed{\lim_{x\to 9}(\sqrt{x}+2) = 5}$$

3 Continuity

Continuity at a number:

A function f is continuous at a number a if

$$\lim_{x\to a} f(x) = f(a).$$

Notice that Definition 1 implicitly requires three things if f is continuous at a:

1. $f(a)$ is defined (that is, a is in the domain of f).
2. $\lim_{x\to a} f(x)$ exists (that is, one-sided limits exist and equal to each other).
3. $\lim_{x\to a} f(x) = f(a)$.

Discontinuity:

$f(x)$ is said to be discontinuous at $x = a$ if

1. $\lim_{x\to a^-} f(x)$ and $\lim_{x\to a^+} f(x)$ exists but are not equal.
2. $\lim_{x\to a^-} f(x)$ and $\lim_{x\to a^+} f(x)$ exists and are equal but not equal to $f(a)$
3. $f(a)$ is not defined.
4. At least one of the one-sided limits doesn't exist.

Types of Discontinuity

Removable: Missing point (a hole) and Isolated point
Non-Removable: Jump, Infinite, Oscillatory

Directional Continuity:

A function f is continuous from the right at a number a if

$$\lim_{x \to a^+} f(x) = f(a)$$

and f is continuous from the left at a if

$$\lim_{x \to a^-} f(x) = f(a)$$

Continuity Test Example

$f(x) = e^{\frac{1}{x}}$ at $x = 0$

$$f(0) \text{ is undefined}$$

$$\lim_{x \to 0^-} e^{\frac{1}{x}} = 0$$

$$\lim_{x \to 0^+} e^{\frac{1}{x}} \to \infty$$

$$\lim_{x \to 0^-} e^{\frac{1}{x}} \neq \lim_{x \to 0^+} e^{\frac{1}{x}} \implies \lim_{x \to 0} e^{\frac{1}{x}} \text{ DNE}$$

$$\boxed{\text{Non-Removable Discontinuity: Infinite}}$$

Continuity on Interval:

A function $f(x)$ is continuous on the open interval (a, b) if it is continuous at every point in the interval (a, b).

A function $f(x)$ is continuous on the closed interval $[a, b]$ if

- (i) It is continuous at every point in the interval (a, b).
- (ii) It is continuous from the right.
- (iii) It is continuous from the left.

Functions attained from the sum, difference, product, and quotient of other functions follow the following continuity principles:

- If $f(x)$ and $g(x)$ are continuous on an interval I, then the functions formed by their sum, difference, product, and quotient are also continuous on I. Note: The quotient $\dfrac{f(x)}{g(x)}$ is continuous only when $g(x) \neq 0$ within the interval.
- If $f(x)$ is continuous and $g(x)$ is discontinuous at $x = a$, then $f(x) \pm g(x)$ is discontinuous at $x = a$.
- If $f(x)$ is continuous and $g(x)$ is discontinuous at $x = a$, then the product function $h(x) = f(x)g(x)$ may or may not be continuous at $x = a$.
- If both functions $f(x)$ and $g(x)$ are discontinuous at $x = a$, then a function formed by applying algebraic operations to $f(x)$ and $g(x)$ may or may not be continuous.
- If $g(x)$ is continuous at $x = a$ and $f(x)$ is continuous at $g(a)$ then the composite function $f(g(x))$ is continuous at $x = a$.

Continuity of Function Involving Limit at Infinity

We know that $\lim_{n\to\infty} a^n = \begin{cases} 0 & 0 \le a < 1 \\ 1 & a = 1 \\ \to \infty & a > 1 \end{cases}$

Also, $\lim_{n\to\infty} a^{2n} = \lim_{n\to\infty} (a^2)^n = \begin{cases} 0 & 0 \le a^2 < 1 \\ 1 & a^2 = 1 \\ \to \infty & a^2 > 1 \end{cases} = \begin{cases} 0 & -1 < a < 1 \\ 1 & a = \pm 1 \\ \to \infty & a < -1 \text{ or } a > 1 \end{cases}$

4 Differentiability

Derivative of a function

Derivative of a function f at a fixed number a:

$$f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h} = \lim_{x\to a} \frac{f(x) - f(a)}{x - a}$$

Derivative as a function:

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

Definition 1

A function f is differentiable at a if $f'(a)$ exists. It is differentiable on an open interval (a, b) [or (a, ∞) or $(-\infty, b)$ or $(-\infty, \infty)$] if it is differentiable at every number in the interval.

Theorem 1 Differentiability implies Continuity

If f is differentiable at a, then f is continuous at a.

Note: The converse of Theorem 1 is false; that is, there are functions that are continuous but not differentiable.

Reasons for Non-Differentiability

Case A: $f(x)$ is continuous at $x = a$ and both $f'(a^+)$ and $f'(a^-)$ exist but are not equal.

Case B: $f(x)$ is discontinuous at $x = a$.

Case C: $f(x)$ is continuous at $x = a$ and either or both $f'(a^+)$ and $f'(a^-)$ are not finite.

Differentiability Test Example

Discuss the differentiability of $f(x) = \begin{cases} \dfrac{\sin x^2}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$ at $x = 0$

$$f'_-(0) = \lim_{h\to 0^-} \frac{\frac{\sin h^2}{h} - 0}{h} = \lim_{h\to 0^-} \frac{\sin h^2}{h^2} = 1$$

$$f'_+(0) = \lim_{h \to 0^+} \frac{\frac{\sin h^2}{h} - 0}{h} = \lim_{h \to 0^+} \frac{\sin h^2}{h^2} = 1$$

$$f'_-(0) = f'_+(0) = 1 \implies f'(0) = 1$$

$$\boxed{f(x) \text{ is differentiable at } x = 0}$$

Functions attained from the sum, difference, product, and quotient of other functions follow the following differentiability principles:

- If $f(x)$ and $g(x)$ are differentiable on the interval I, then any function formed by their arithmetic combination—sum, difference, product, or quotient—is likewise differentiable and continuous on I, provided the denominator of the quotient is non-zero.
- If a function $f(x)$ is differentiable at a point but $g(x)$ is not, their sum or difference $f(x) \pm g(x)$ will inherit the non-differentiability of $g(x)$ at that same point.
- If a function $f(x)$ is differentiable and $g(x)$ is not differentiable at $x = a$, then the product function $f(x)g(x)$ may or may not be differentiable at $x = a$.
- When two functions $f(x)$ and $g(x)$ are both non-differentiable at $x = a$, their sum, difference, product, or quotient may result in a function that is either differentiable or non-differentiable at that point.

5 Derivative Rules

Constant Times a Function

$$(cf(x))' = cf'(x) \quad \text{(Euler/Lagrange)}$$

$$\frac{d}{dx}(cf(x)) = c\frac{df(x)}{dx} \quad \text{(Leibniz)}$$

Sum/Difference Rule

$$(f(x) \pm g(x))' = f'(x) \pm g'(x)$$

$$\frac{d}{dx}(f(x) \pm g(x)) = \frac{d}{dx}f(x) \pm \frac{d}{dx}g(x)$$

Product Rule

$$(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$$

$$\frac{d}{dx}(fg) = f'g + fg'$$

Quotient Rule

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

Power Rule

For any real number $n \neq 0$, the derivative of x^n with respect to x is given by

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

The Chain Rule

$$(f(g(x)))' = f'(g(x)) \cdot g'(x)$$

Differentiate the outside, leave the inside alone, then multiply by the derivative of the inside!

Trigonometric Derivatives

$$\frac{d}{dx}(\sin x) = \cos x, \quad \frac{d}{dx}(\cos x) = -\sin x,$$
$$\frac{d}{dx}(\tan x) = \sec^2 x, \quad \frac{d}{dx}(\cot x) = -\csc^2 x,$$
$$\frac{d}{dx}(\sec x) = \sec x \tan x, \quad \frac{d}{dx}(\csc x) = -\csc x \cot x.$$

Exponential Derivatives

$$\frac{d}{dx}(e^x) = e^x,$$
$$\frac{d}{dx}(a^x) = a^x \ln a \quad (a > 0, a \neq 1).$$

Logarithmic Derivatives

$$\frac{d}{dx}(\ln x) = \frac{1}{x} \quad (x > 0),$$
$$\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a} \quad (a > 0, a \neq 1).$$

Implicit Differentiation

Implicit differentiation makes use of the Chain rule to differentiate a function which cannot be explicitly expressed in the form $y = f(x)$. Such a function is known as an implicitly defined function. By the Chain rule, the derivative with respect to x of a function $f(y(x))$ is given by:

$$\frac{d}{dx}(f(y)) = f'(y) \cdot y' \quad \text{or} \quad \frac{d}{dx}(f(y)) = f'(y) \cdot \frac{dy}{dx}$$

Logarithmic Differentiation

Let $y = f(x)^{g(x)}$. Then $\ln y = g(x)\ln f(x)$. Differentiating with respect to x, yields

$$\frac{1}{y}y' = g'(x)\ln f(x) + g(x)\frac{f'(x)}{f(x)}.$$

Let $y = \dfrac{f_1(x)\cdot f_2(x)\cdot f_3(x)\cdots}{g_1(x)\cdot g_2(x)\cdot g_3(x)\cdots}$. Then

$$\ln y = \ln(f_1(x)\cdot f_2(x)\cdot f_3(x)\cdots) - \ln(g_1(x)\cdot g_2(x)\cdot g_3(x)\cdots)$$

$$\frac{1}{y}y' = \left[\frac{(f_1(x))'}{f_1(x)} + \frac{(f_2(x))'}{f_2(x)} + \frac{(f_3(x))'}{f_3(x)} + \cdots\right] - \left[\frac{(g_1(x))'}{g_1(x)} + \frac{(g_2(x))'}{g_2(x)} + \frac{(g_3(x))'}{g_3(x)} + \cdots\right].$$

6 Existence Theorem

6.1 Intermediate Value Theorem

The Intermediate Value Theorem (IVT):

Suppose that f is continuous on the closed interval $[a, b]$ and let N be any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then there exists a number c in (a, b) such that $f(c) = N$.

IVT Example

$f(x) = x^3 - 3x - 19$ or $x^3 - 3x - 19 = 0$ on interval $[1, 5] \implies a = 1, b = 5$ s

1. $f(x)$ is a continuous polynomial function on $[1, 5]$.
2. $f(1) = -21, f(5) = 91$
$$-21 < N < 91 \implies \text{let } N = 0$$
3. By the Intermediate Value Theorem, since $f(x)$ is continuous on $[1, 5]$, and $f(1) < 0 < f(5)$, then there exist a number $c \in (1, 5)$ such that $f(c) = 0$. i.e. $c^3 - 3c - 19 = 0$ is solvable on $(1, 5)$.

6.2 Extreme Value Theorem

Absolute Extrema

A function f has an absolute maximum (or global maximum) at c if $f(c) \geq f(x)$ for all x in the domain of f. The number $f(c)$ is called the maximum value of f. Similarly, f has an absolute minimum at c if $f(c) \leq f(x)$ for all x in the domain of f and the number $f(c)$ is called the minimum value of f. The maximum and minimum values of f are called the extreme values of f.

Local (Relative) Extrema

A function f has a local maximum (or relative maximum) at c if $f(c) \geq f(x)$ when x is near c. [This means that $f(c) \geq f(x)$ for all x in some open interval containing c.] Similarly, has f a local minimum at c if $f(c) \leq f(x)$ when x is near c.

The Extreme Value Theorem (EVT)

If f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some numbers c and d in $[a, b]$.

The Fermat's Theorem

If f has a local maximum or minimum at c, and if $f'(c)$ exists, then $f'(c) = 0$.

Critical Number

A critical number of a function f is a number c in the domain of f such that either $f'(c) = 0$ or $f'(c)$ does not exist.

The Closed Interval Method

To find the absolute maximum and minimum values of a continuous function f on a closed interval $[a, b]$:

1. Find the values of f at the critical numbers of f in (a, b).
2. Find the values of f at the endpoints of the interval.
3. The largest of the values from Steps 1 and 2 is the absolute maximum value; the smallest of these values is the absolute minimum value.

Closed Interval Method Example

For the following function, find the maximum and minimum values on the indicated intervals, by finding the critical points, and comparing the values at these points with the values at the endpoints.

$f(x) = 3x^4 - 6x^3 + 6x^2$ on $\left[-\frac{1}{2}, \frac{1}{2}\right]$

$$f'(x) = 12x^3 - 18x^2 + 12x$$
$$6x(2x^2 - 3x + 2) = 0 \implies f'(0) = 0$$
$$f(0) = 0$$
$$f(-\frac{1}{2}) = \frac{39}{16}$$
$$f(\frac{1}{2}) = \frac{15}{16}$$
$$\boxed{\text{max} : (-\frac{1}{2}, \frac{39}{16}), \text{min} : (0, 0)}$$

6.3 Mean Value Theorem

Rolle's Theorem

Let f be a function that satisfies the following three hypotheses:

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b).
3. $f(a) = f(b)$

Then there is a number c in (a, b) such that $f'(c) = 0$.

The Mean Value Theorem

Let f be a function that satisfies the following hypotheses:

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b).

Then there is a number c in (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Theorem

If $f'(x) = 0$ for all x in an interval (a, b), then f is constant on (a, b).

Corollary

If $f'(x) = g'(x)$ for all x in an interval (a, b), then $f - g$ is constant on (a, b); that is, $f(x) = g(x) + c$ where c is a constant.

Rolle's Thoerem Example

Verify that the function satisfies the three hypotheses of Rolle's Theorem on the given interval. Then find all numbers that satisfy the conclusion of Rolle's Theorem.

$f(x) = \sqrt{x} - \frac{x}{3}, \quad [0, 9]$

1. $f(x)$ is continuous on $[0, 9]$ as $f(x)$ is constructed by polynomial functions and $x \geq 0$.
2. $f(x)$ is differentiable on $(0, 9)$.
3. $\begin{cases} f(0) = \sqrt{0} - \frac{0}{3} = 0 \\ f(9) = \sqrt{9} - \frac{9}{3} = 0 \end{cases} \implies f(0) = f(9)$

By Rolle's Theorem,

$$\boxed{\exists c \in (0, 9) \text{ s.t. } f'(c) = 0}$$

$$f'(c) = \frac{1}{2\sqrt{c}} - \frac{1}{3} = 0$$
$$2\sqrt{c} = 3$$
$$\boxed{c = \frac{9}{4}}$$

MVT Example

Verify that the function satisfies the hypotheses of the Mean Value Theorem on the given interval. Then find all numbers c that satisfy the conclusion of the Mean Value Theorem.

$f(x) = 3x^2 + 2x + 5, \quad [-1, 1]$

1. $f(x)$ is continuous on $[-1, 1]$ as $f(x)$ is constructed by polynomial functions.
2. $f(x)$ is differentiable on $(-1, 1)$.

By the Mean Value Theorem,

$$\boxed{\exists c \in (-1, 1) \text{ s.t. } f'(c) = 2}$$
$$f'(c) = 6c + 2 = 2$$
$$\boxed{c = 0}$$

7 Derivatives with Function Graph

Increasing/Decreasing Test

- If $f'(x) > 0$ on an interval, then f is increasing on that interval.
- If $f'(x) < 0$ on an interval, then f is decreasing on that interval.

The First Derivative Test

Suppose that c is a critical number of a continuous function f:

- If f' changes from positive to negative at c, then f has a local maximum at c.
- If f' changes from negative to positive at c, then f has a local minimum at c.
- If f' doesn't change sign at c, then f has no local maximum or minimum at c.

Definition of Concavity

If the graph of f lies above all of its tangents on an interval I, then it is called concave upward on I. If the graph of f lies below all of its tangents on an interval I, then it is called concave downward on I.

Concavity Test

- If $f''(x) > 0$ for all x in I, then the graph of f is concave upward in I.
- If $f''(x) < 0$ for all x in I, then the graph of f is concave downward in I.

Inflection Point

A point P on a curve $y = f(x)$ is called an inflection point if f is continuous there and the curve changes from concave upward to concave downward or from concave downward to concave upward.

The Second Derivative Test

Suppose f'' is continuous near c. Then:

- If $f'(c) = 0$ and $f''(c) > 0$ then f has a local minimum at c.
- If $f'(c) = 0$ and $f''(c) < 0$ then f has a local maximum at c.

Function Analysis Example

$f(x) = \dfrac{\ln x}{\sqrt{x}}$

$$x > 0$$

$$f'(x) = \frac{\frac{\sqrt{x}}{x} - \frac{\ln x}{2\sqrt{x}}}{x} = \frac{2 - \ln x}{2x\sqrt{x}}$$

$$f'(x) = 0 \implies \ln x = 2 \implies x = e^2$$

$$f'(x) > 0, \quad \forall 0 < x < e^2$$

$$f'(x) < 0, \quad \forall x > e^2$$

$f(x)$ is increasing on $(0, e^2)$ because $f'(x) > 0$ on this interval.

$f(x)$ is decreasing on (e^2, ∞) because $f'(x) < 0$ on this interval.

$f'(x)$ changes from positive to negative at $x = e^2$. Therefore, $f(x)$ has a local maximum at $x = e^2$.

$$f(e^2) = \frac{\ln e^2}{\sqrt{e^2}} = \frac{2}{e}$$

$$\boxed{\max : \frac{2}{e}}$$

$$f''(x) = \frac{(\ln x - 2)3\sqrt{x} - 2\sqrt{x}}{4x^3}$$

$$f''(x) = 0 \implies 3\ln x = 8 \implies x = e^{\frac{8}{3}}$$

$$f''(x) > 0, \quad \forall x > e^{\frac{8}{3}}$$

$$f''(x) < 0, \quad \forall 0 < x < e^{\frac{8}{3}}$$

$f(x)$ is concave up on $(e^{\frac{8}{3}}, \infty)$ because $f''(x) > 0$ on this interval.

$f(x)$ is concave down on $(0, e^{\frac{8}{3}})$ because $f''(x) < 0$ on this interval.

Inflection point: $\boxed{x = e^{\frac{8}{3}}}$